Ridge Regression

Ridge regression is a supervised learning algorithm used for predicting a continuous target variable. It is an extension of linear regression that adds a regularization term to the loss function. This regularization term helps reduce overfitting and makes the model more stable when the feature matrix contains highly correlated features.

Suppose we are given a dataset consisting of n observations and p features. Let

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

be the feature matrix and

$$\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

be the vector of target values.

As in linear regression, after absorbing the bias term into the feature matrix, the prediction function can be written as

$$\hat{\boldsymbol{y}} = \mathbf{X}\boldsymbol{w},$$

where $\mathbf{X} \in \mathbb{R}^{n\times p}$ is the feature matrix, $\boldsymbol{w} \in \mathbb{R}^p$ is the vector of model coefficients, and $\hat{\boldsymbol{y}} \in \mathbb{R}^n$ is the vector of predicted target values.

In ordinary linear regression, the loss function is

$$L(\boldsymbol{w}) = \frac{1}{2n} \left\|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\right\|^2.$$

Ridge regression adds an L_2 regularization term to penalize large model coefficients. Therefore, the ridge regression loss function is

$$L(\boldsymbol{w}) = \frac{1}{2n} \left\| \mathbf{X}\boldsymbol{w} - \boldsymbol{y} \right\|^2 + \frac{\lambda}{2} \left\| \boldsymbol{w} \right\|^2,$$

where $\lambda \geq 0$ is the regularization parameter. A larger value of λ applies a stronger penalty to large coefficients.

Equivalently, the loss function can be written as

$$L(\boldsymbol{w}) = \frac{1}{2n}(\mathbf{X}\boldsymbol{w} - \boldsymbol{y})^\top(\mathbf{X}\boldsymbol{w} - \boldsymbol{y}) + \frac{\lambda}{2}\boldsymbol{w}^\top\boldsymbol{w}.$$

Taking the derivative with respect to $\boldsymbol{w}$, we obtain

$$\begin{aligned} \frac{\partial L(\boldsymbol{w})}{\partial \boldsymbol{w}} &= \frac{\partial}{\partial \boldsymbol{w}}\left[\frac{1}{2n}(\mathbf{X}\boldsymbol{w} - \boldsymbol{y})^\top(\mathbf{X}\boldsymbol{w} - \boldsymbol{y}) + \frac{\lambda}{2}\boldsymbol{w}^\top\boldsymbol{w}\right] \\ &= \frac{1}{n}\mathbf{X}^\top(\mathbf{X}\boldsymbol{w} - \boldsymbol{y}) + \lambda\boldsymbol{w}. \end{aligned}$$

Setting the derivative equal to $\mathbf{0}$ gives

$$\begin{aligned} \frac{1}{n}\mathbf{X}^\top(\mathbf{X}\boldsymbol{w} - \boldsymbol{y}) + \lambda\boldsymbol{w} &= \mathbf{0} \\ \Leftrightarrow \mathbf{X}^\top(\mathbf{X}\boldsymbol{w} - \boldsymbol{y}) + n\lambda\boldsymbol{w} &= \mathbf{0} \\ \Leftrightarrow \mathbf{X}^\top\mathbf{X}\boldsymbol{w} - \mathbf{X}^\top\boldsymbol{y} + n\lambda\boldsymbol{w} &= \mathbf{0} \\ \Leftrightarrow \left(\mathbf{X}^\top\mathbf{X} + n\lambda\mathbf{I}\right)\boldsymbol{w} &= \mathbf{X}^\top\boldsymbol{y}. \end{aligned}$$

Assuming $\mathbf{X}^\top\mathbf{X} + n\lambda\mathbf{I}$ is invertible, we obtain

$$\boldsymbol{w} = \left(\mathbf{X}^\top\mathbf{X} + n\lambda\mathbf{I}\right)^{-1}\mathbf{X}^\top\boldsymbol{y}.$$

This is the closed-form solution for ridge regression.

When $\lambda = 0$, ridge regression becomes ordinary linear regression. When λ increases, the model coefficients are pushed closer to 0, which can reduce overfitting.